

Exploratory Data Analysis: Unveiling Trends in Food Delivery

Team members:

1. Adithya Reddy Bollaram
2. Nikhil Kattaguri
3. Sathvika Shetti

Title: Unveiling Trends in Food Delivery

Analyzing factors affecting food delivery performance.

Abstract:

We created a predictive model to forecast food delivery times more accurately. We cleaned and analyzed a Kaggle dataset to identify factors that impact delivery performance, such as vehicle type, weather, and traffic. The resulting model provides valuable insights to improve operational efficiency and customer satisfaction in the food delivery industry.

Introduction:

In the fast-paced world of online food delivery, the efficiency of service delivery directly impacts customer satisfaction and business success. Recognizing the need for enhanced predictability and speed in food delivery, A comprehensive analysis titled "Food Delivery Time Analysis." This project is dedicated to dissecting historical data from an extensive dataset provided by an online retail platform sourced from Kaggle. The primary objective is to refine the accuracy of delivery time forecasts, which is crucial for improving both the delivery process and the overall customer experience. This project involves collecting and loading data into an analytical framework with Pandas, followed by data preparation and cleaning to ensure accuracy.

We analyze the data to uncover variables that impact delivery times. This includes examining order type, delivery personnel details, and external factors like traffic and weather. We use visual techniques to highlight patterns and trends that affect delivery performance. We created a predictive model to improve delivery services. It considers various factors including vehicle types, weather conditions, city dynamics, and festive periods. The goal is to predict delivery times accurately and adapt to different delivery environments.

Methodology

Our methodology for the "Food Delivery Time Analysis" project included several key steps:

1. **Data Collection:** We gathered detailed delivery data from Kaggle.
2. **Data Cleaning:** We cleaned the data by removing duplicates, filling in missing values, and eliminating irrelevant information.
3. **Data Exploration and Visualization:** Using Python's Pandas and visualization libraries, we analyzed and visualized the data to identify key factors affecting delivery times.

4. **Predictive Modeling:** We developed a regression model to predict delivery times based on variables like weather, traffic, and vehicle type.

DATA LOADING

Loading the dataset into a data frame using Pandas

```
In [75]: df = pd.read_csv("food_delivery.csv")
```

Drop unnecessary columns

```
In [8]: # Drop Id & Delivery_person_ID columns
df.drop(["ID", "Delivery_person_ID"], axis=1, inplace=True)
df.shape
```

```
Out[8]: (45593, 19)
```

```
In [10]: def update_datatypes(data):
    data["Delivery_person_Age"] = data["Delivery_person_Age"].astype("float64")
    data["Delivery_person_Ratings"] = data["Delivery_person_Ratings"].astype("float64")
    data["multiple_deliveries"] = data["multiple_deliveries"].astype("float64")
    data["Order_Date"] = pd.to_datetime(data["Order_Date"], format="%d-%m-%Y")
    data["Time_Orderd"] = pd.to_timedelta(data["Time_Orderd"])
    data["Time_Order_picked"] = pd.to_timedelta(data["Time_Order_picked"])
    return data
```

```
df = update_datatypes(df)
df.info()
```

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 45593 entries, 0 to 45592
Data columns (total 19 columns):
 #   Column                                Non-Null Count  Dtype
---  ---                                -
 0   Delivery_person_Age                   43739 non-null  float64
 1   Delivery_person_Ratings               43685 non-null  float64
 2   Restaurant_latitude                   45593 non-null  float64
 3   Restaurant_longitude                 45593 non-null  float64
 4   Delivery_location_latitude            45593 non-null  float64
 5   Delivery_location_longitude           45593 non-null  float64
 6   Order_Date                           45593 non-null  datetime64[ns]
 7   Time_Orderd                          43862 non-null  timedelta64[ns]
 8   Time_Order_picked                    45593 non-null  timedelta64[ns]
 9   Weatherconditions                    44977 non-null  object
10   Road_traffic_density                  44992 non-null  object
11   Vehicle_condition                    45593 non-null  int64
12   Type_of_order                        45593 non-null  object
13   Type_of_vehicle                      45593 non-null  object
14   multiple_deliveries                  44600 non-null  float64
15   Festival                             45365 non-null  object
16   City                                 44393 non-null  object
17   Time_taken(min)                      45593 non-null  int64
18   Code                                 45593 non-null  object
dtypes: datetime64[ns](1), float64(7), int64(2), object(7), timedelta64[ns](2)
memory usage: 6.6+ MB
```

Exploring the number of rows & columns, ranges of values etc.

```
In [3]: df.head()
Out[3]:
```

	ID	Delivery_person_ID	Delivery_person_Age	Delivery_person_Ratings	Restaurant_latitude	Restaurant_longitude	Delivery_location_latitude	Delivery
0	0x4607	INDORES13DEL02	37	4.9	22.745049	75.892471	22.765049	
1	0xb379	BANGRES18DEL02	34	4.5	12.913041	77.683237	13.043041	
2	0x5d6d	BANGRES19DEL01	23	4.4	12.914264	77.678400	12.924264	
3	0x7a6a	COIMBRES13DEL02	38	4.7	11.003669	76.976494	11.053669	
4	0x70a2	CHENRES12DEL01	32	4.6	12.972793	80.249982	13.012793	

```
In [4]: df.shape
Out[4]: (45593, 20)
```

DATA CLEANING

```
In [11]: #Checking Null Values
df.isnull().sum().sort_values(ascending = False)[:10]

Out[11]: Delivery_person_Ratings    1908
Delivery_person_Age                1854
Time_Orderd                        1731
City                              1200
multiple_deliveries                993
Weatherconditions                  616
Road_traffic_density              601
Festival                          228
Order_Date                        0
Delivery_location_longitude        0
dtype: int64

In [12]: total = df.isnull().sum().sort_values(ascending=False)
percent = df.isnull().mean()*100
missing_data = pd.concat([total, percent], axis=1, keys=['Total', 'Percent'])
missing_data.head(10)

Out[12]:
```

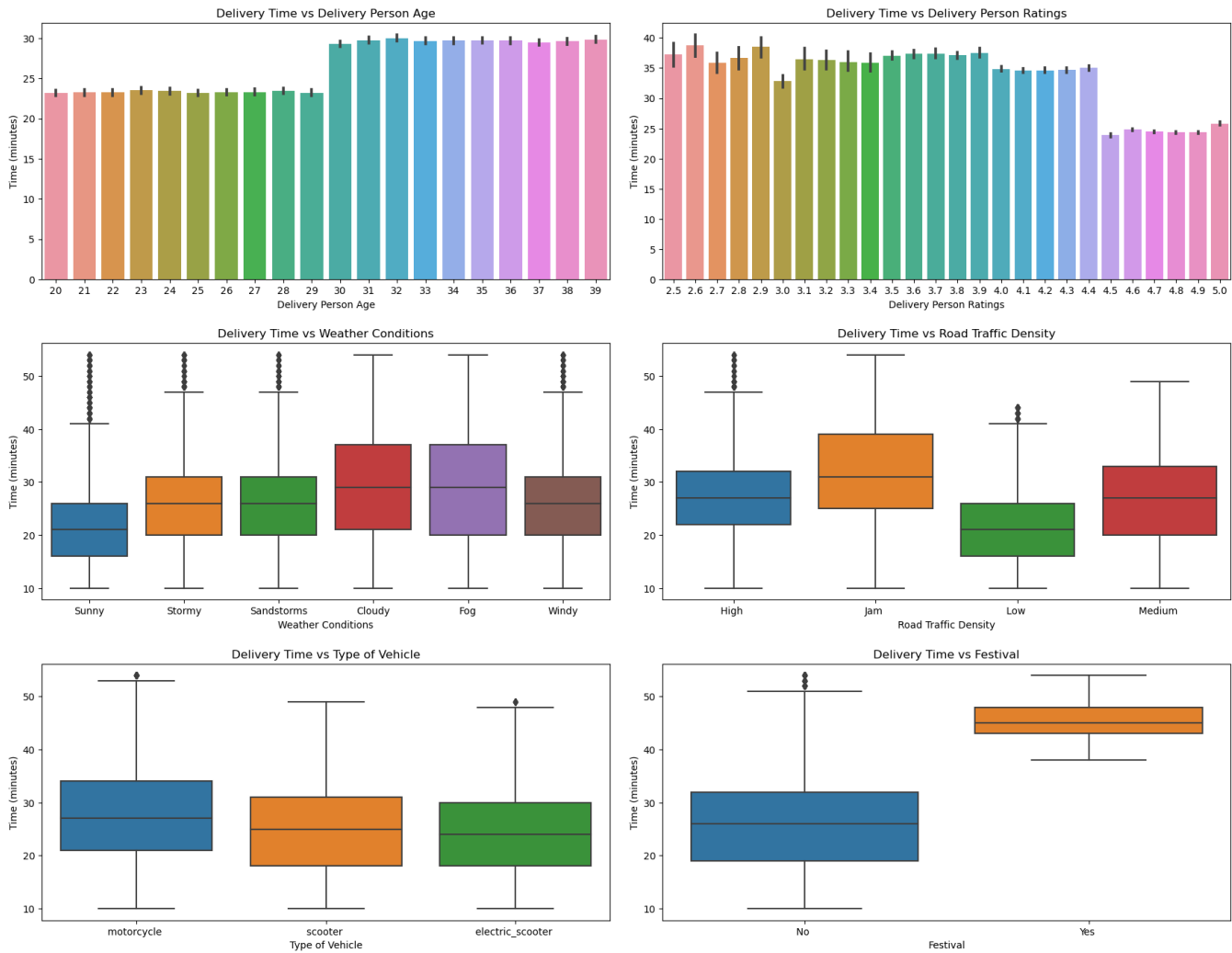
	Total	Percent
Delivery_person_Ratings	1908	4.184853
Delivery_person_Age	1854	4.066414
Time_Orderd	1731	3.796635
City	1200	2.631983
multiple_deliveries	993	2.177966
Weatherconditions	616	1.351085
Road_traffic_density	601	1.318185
Festival	228	0.500077
Order_Date	0	0.000000
Delivery_location_longitude	0	0.000000

Results:

Data Visualization:

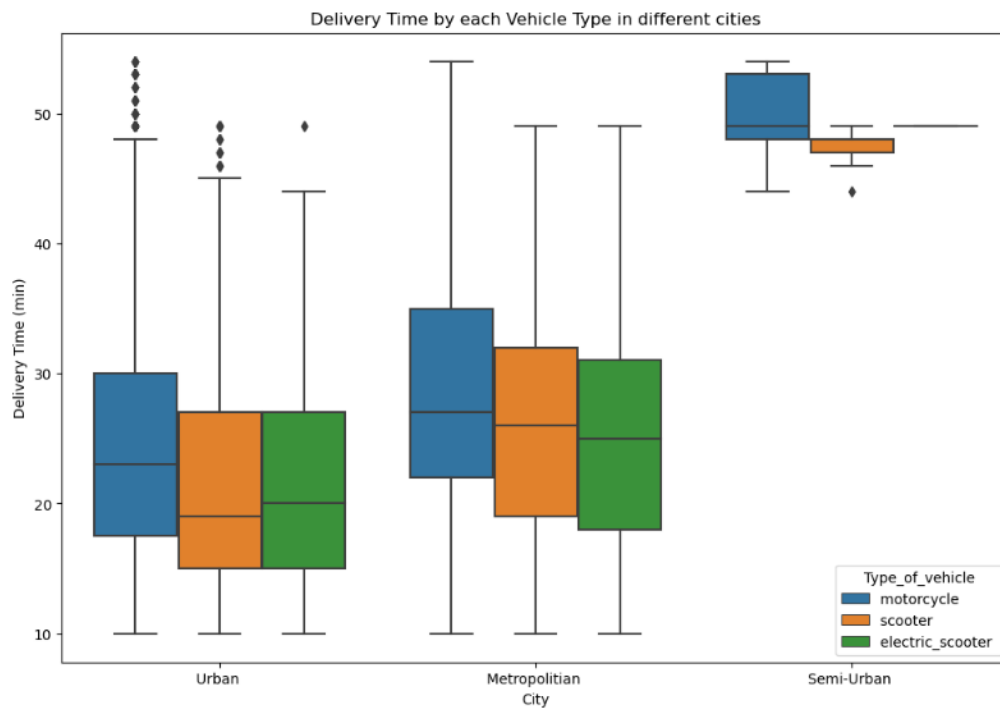
Various visualization techniques are employed to illustrate the relationships and distributions within the data.

Can we build a predictive model to estimate delivery time based on a combination of weather conditions, road traffic density, and the type of vehicle used, while also considering the festival?

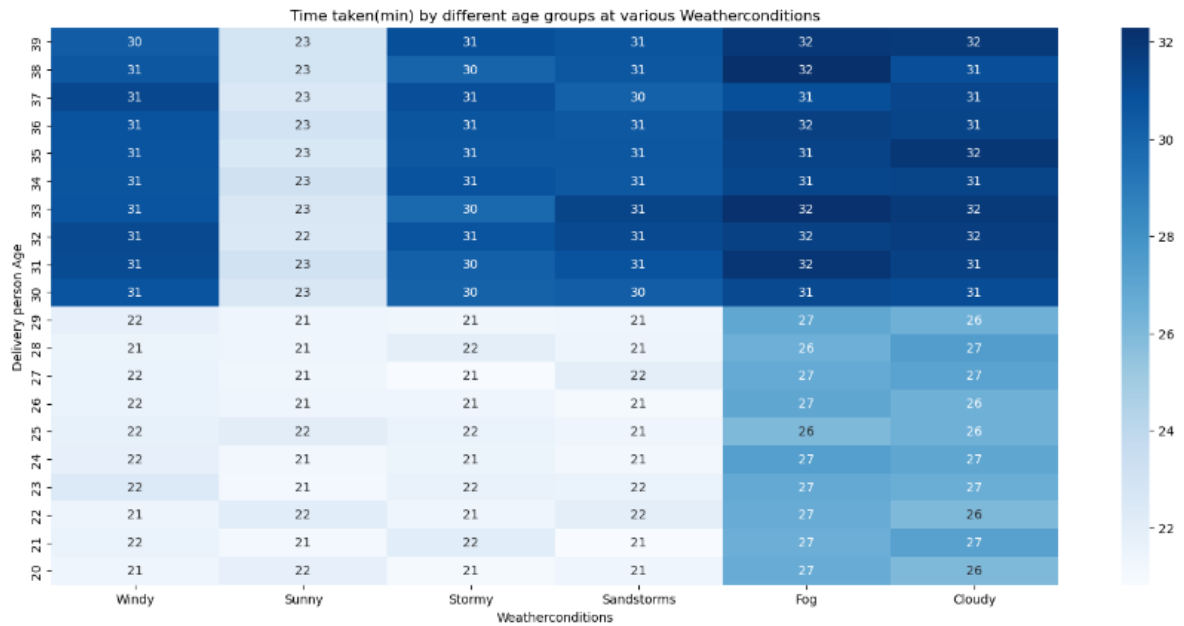


Delivery Time taken by each Vehicle Type in different cities

```
n [18]: # Boxplot of delivery time by city and type of vehicle
plt.figure(figsize=(12, 8))
sns.boxplot(x='City', y='Time_taken(min)', data=df, hue='Type_of_vehicle')
plt.title('Delivery Time by each Vehicle Type in different cities')
plt.xlabel('City')
plt.ylabel('Delivery Time (min)')
plt.show()
```



```
In [27]: # Visualize as heatmap
plt.figure(figsize=(18, 8))
sns.heatmap(pivot1, annot=True, cmap="Blues")
plt.title('Time taken(min) by different age groups at various Weatherconditions')
plt.xlabel('Weatherconditions')
plt.ylabel('Delivery person Age')
plt.gca().invert_xaxis()
plt.gca().invert_yaxis()
plt.show()
```

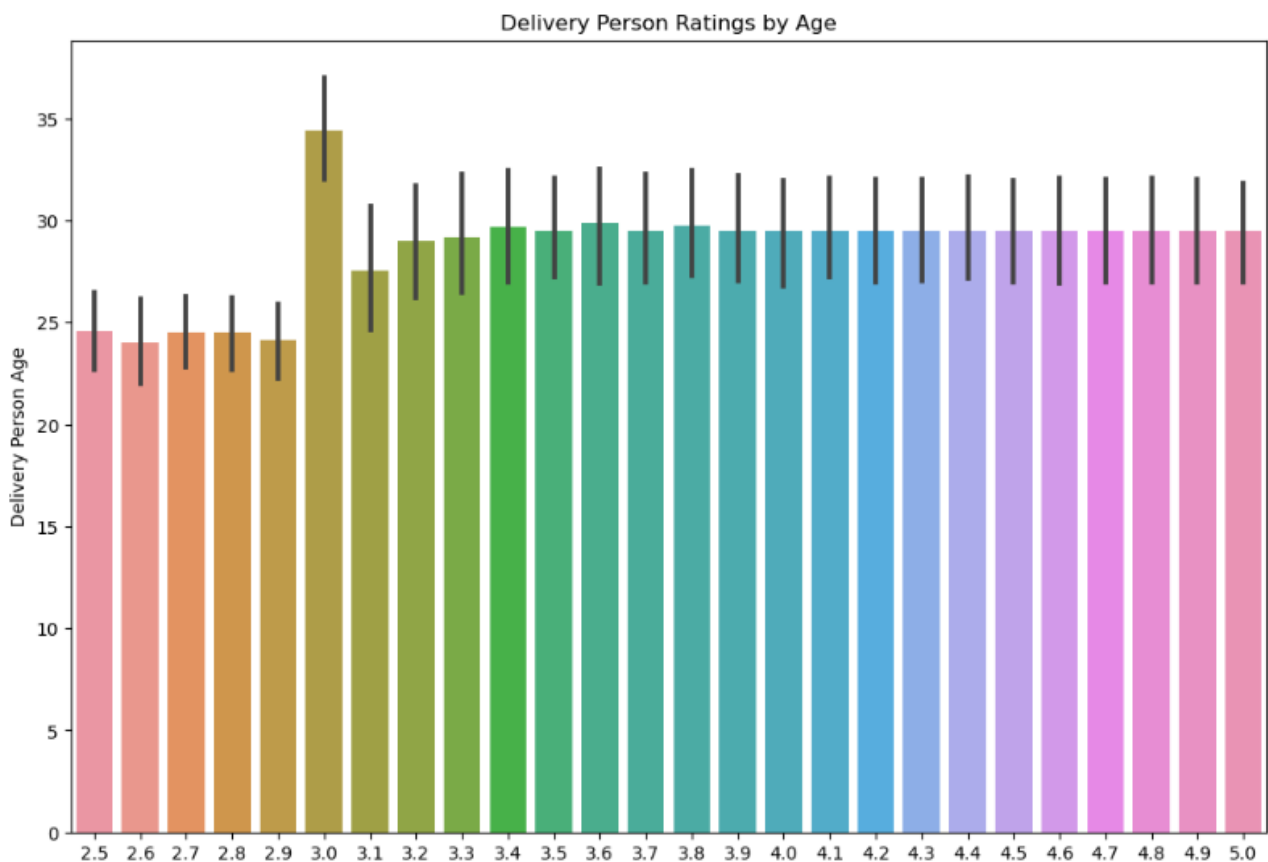


Finding relationship between Delivery person age and ratings

- Create new columns, merge multiple dataset and perform grouping/aggregation

```
# Group the data by 'Delivery_person_Age' and 'Delivery_person_Ratings'
age_ratings_group = df.groupby(['Delivery_person_Age', 'Delivery_person_Ratings'])['Time_taken(min)'].mean().reset_index()

# Create a bar plot
plt.figure(figsize=(12, 8))
sns.barplot(x='Delivery_person_Ratings', y='Delivery_person_Age', data=age_ratings_group)
plt.title('Delivery Person Ratings by Age')
plt.xlabel('Delivery Person Ratings')
plt.ylabel('Delivery Person Age')
plt.show()
```



The analysis of the food delivery data revealed several key insights:

Delivery Efficiency: Vehicle type and traffic conditions significantly influenced delivery times. Motorcycles were generally faster in urban settings due to their ability to navigate through traffic.

Impact of Environmental Factors: Weather conditions played a crucial role, with poor weather typically extending delivery times.

Conclusion:

Our analysis effectively utilized data to forecast food delivery times. We identified critical factors such as vehicle type, weather, and traffic that impact the delivery speed. These insights will help improve delivery efficiency and customer satisfaction. The project's findings offer valuable strategies for food delivery companies to enhance their operations.

Future Scope:

In the future, we plan to improve this project by integrating real-time data on traffic and weather. We also aim to explore emerging technologies such as AI route optimization and drone deliveries. To make the model more accurate and applicable, we will expand the study to include more diverse geographic data. These advancements could significantly enhance our predictive capabilities and ensure that delivery services remain innovative and customer-focused.

References:

- [1] Stack Overflow. (2019). Merge, join, concatenate, and compare.
<https://stackoverflow.com/questions/53645882/pandas-merging-101>
- [2] Pandas Tutorial:
<https://www.w3schools.com/python/pandas/default.asp>
- [3] Kaggle:
<https://www.kaggle.com/>